INCH-POUND

MIL-B-51254(MU)
AMENDMENT 5
19 April 1991
SUPERSEDING
AMENDMENT 4
24 August 1988

MILITARY SPECIFICATION

BREATHING APPARATUS, OXYGEN GENERATING, M20

This amendment forms a part of MIL-B-51254(MU), dated 12 December 1966, and is approved for use by the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1:

Under "SPECIFICATIONS" "FEDERAL", delete the following:

"NN-P-00515 - Plywood, Container Grade.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material)."

Under "SPECIFICATIONS" "FEDERAL", add the following:

"L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin.
PPP-B-26 - Bag, Plastic, Polyethylene.
PPP-T-60 - Tape: Pressure-Sensitive, Adhesive, Waterproof, For Packaging.
PPP-F-320 - Fiberboard; Corrugated and Solid Sheet Stock (Container Grade); and Cut Shapes.

AMSC N/A FSC 4240

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QQ-S-781 - Strapping, Steel, Flat and Seals."

Under "SPECIFICATIONS" "MILITARY", delete the following:

"MIL-P-116 - Preservation, Methods of."

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2.1:

Under "PUBLICATIONS" "DEPARTMENT OF THE ARMY TECHNICAL MANUALS":

Delete "TM 3-4240-212-15" and "TM 3-4240-212-25P" and their titles and substitute "TM 3-4240-212-14P - Operator's Organizational, Direct Support, and General Support Maintenance Manual (Including Repair Parts and Special Tools List) for Breathing Apparatus, Oxygen-Generating, M20."

At the end of the paragraph add the following:

"CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 199 - Hazardous Materials Regulations

(The Code of Federal Regulations and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)"

2.2: Delete in its entirety and substitute the following:

"**2.2 Non-Government publications.** The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

INTERNATIONAL CIVIL AVIATION ORGANIZATION

Technical Instructions for the Safe Transport of Dangerous Goods by Air

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(Application for copies should be addressed to the International Civil Aviation Organization, 1000 Sherbrooke Street West, Suite 400, Montreal, Quebec, Canada H3A 2R2.)

INTERNATIONAL MARITIME ORGANIZATION

International Maritime Dangerous Goods Code

(Application for copies should be addressed to the International Maritime Organization, 101-104 Piccadilly, London, WIV OAE, England.)"

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3.1.7: Delete in its entirety and substitute the following:

"**3.1.7 Carrying case design and construction.** A carrying case, conforming to 4A1W steel box in accordance with International Civil Aviation Organization - Technical Instructions for the Safe Transport of Dangerous Goods by Air (ICAO-TDGA) or International Maritime Organization - International Maritime Dangerous Goods Code (IMO-IMDG), shall be provided to retain the apparatus, a spanner wrench to couple and uncouple the hoses, three type A-3 canisters (see 3.1.9), instruction booklet or card, an antifogging kit (see 3.1.11) and a hood (see 3.1.12). Unless otherwise specified in the contract or order, the color of the carrying case shall be green No. 34087 of FED-STD-595. If the case is provided with any metal parts, they shall be a corrosion resistant metal or metal parts treated with standard commercial treatment for corrosion resistance prior to painting."

3.1.9: Delete in its entirety and substitute the following:

"**3.1.9 Canisters.** Each apparatus shall be provided with three type A-3 canisters conforming to MIL-C-17671."

3.1.10, lines 1 and 2: Delete "each of TM 3-4240-212-15 and TM 3-4240-212-25P" and substitute "of TM 3-4240-212-14P".

3.1.12: Delete in its entirety and substitute the following:

"**3.1.12 Hood.** One hood supplied as an accessory shall be Mine Safety Appliance Co. part no D304157."

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4.2.1, line 1: Add "packaged" before "preproduction".

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4.2.2.1: Delete in its entirety and substitute the following:

"**4.2.2.1 For examination.** Packaged sample apparatus shall be examined in accordance with the classification of defects (4.3.3.3(b)). After completion of the packaging inspection, the sample apparatus shall then be examined in accordance with the classification of defects (4.3.3.3(a))."

4.2.2.2, line 1: Add "packaged" before "apparatus".

4.3.2.1: Delete in its entirety and substitute the following:

"**4.3.2.1 For examination and nondestructive tests.** Sampling for examination and test of item, and the packaging thereof, shall be conducted in accordance with MIL-STD-105."

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4.3.3.1: Delete in its entirety and substitute the following:

"**4.3.3.1 Examination and nondestructive tests.** Sample apparatus, unit packed in accordance with 5.1, shall be examined and tested in accordance with defects listed in 4.3.3.3(b) and MIL-STD-105. After completion of the packaging inspection, the sample apparatus shall then be examined and tested in accordance with the classification of defects (4.3.3.3(a)) and MIL-STD-105. MIL-STD-289 and MIL-STD-407 shall apply as visual inspection guides for rubber sheet material and rubber molded items."

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4.3.3.3(b):

Change title to read "Packaging inspection (section 5)."

Under Major Defects, delete text in its entirety and substitute the following:

"101	Packing container not as specified (see 3.1.7)
102	Overpack not as specified or damaged
103	Closure of pack or overpack not as specified
104	Carrying case not enclosed in polyethylene wrap or bag as specified
105	Cushioning, blocking or bracing missing or insufficient to make a tight package
106	Marking missing, incorrect or illegible"

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4.3.4.1(b), lines 2, 3, and 4: Delete "oxygen supply" and substitute "conditioned compressed air supply".

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4.3.4.3:

Line 4: Delete "oxygen supply" and substitute "conditioned compressed air supply".

Lines 6, 10, and 12: Delete "the oxygen source" and substitute "the air source".

PAGES 9 and 10

Section 5, delete in its entirety and substitute the following:

"5. PACKAGING

5.1 Packing, level A. No preservative compounds or oils shall be applied to any portion of the breathing apparatus. The breathing apparatus, consisting of a facepiece and hose assembly, back-carry harness assembly, breathing bag assembly, anti-fogging kit, hood, canisters, spanner wrench and technical publications shall be placed inside the carrying case (see 3.1.7) in such a manner to prevent damage to the rubber components of the breathing apparatus. Internal cushioning, blocking, or bracing shall be used as necessary to prevent movement of the breathing apparatus or components within the carrying case. The cover shall be secured with the fasteners provided. The case shall be either enclosed in a wrap of polyethylene conforming to L-P-378, type II (folded ends taped with tape conforming to PPP-T-60, type IV), or in a polyethylene bag conforming to PPP-B-26, type II, style 1 or 2, 0.004 inch thick material. The bag, if used, shall be closed as follows: Style 1 bags shall be either heat sealed in accordance with the bag manufacturers recommendation, or the open end taped with tape conforming to PPP-T-60, type IV. Style 2 bags shall be closed with wire tie provided.

5.1.1 Performance oriented packaging. The contractor shall be responsible for testing and marking the pack (shipping container) in accordance with the applicable packaging performance tests specified in 49 CFR 171 to 199 and ICAO-TDGA. The contractor shall furnish the Government with a certificate and test report verifying the successful completion of these tests.

5.2 Overpacking. Level shall be as specified (see 6.2).

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5.2.1 Level A. The breathing apparatus, unit packed as specified in 5.1, shall be packed in an overseas type, style optional, class A, cleated plywood box conforming to PPP-B-601. The box shall be minimum cube to contain the unit packed breathing apparatus. Filler pads conforming to PPP-F-320, grade W5c or W6c, shall be used as necessary to ensure a tight pack. The box shall be closed and strapped in accordance with the appendix to PPP-B-601. Strapping and seals shall conform to QQ-S-781 (Strapping - type I (Regular Duty), finish B, grade 2 and Seals - type D, styles I or IV, finish B, grade 2).

5.2.2 Level B. The breathing apparatus, unit packed as specified in 5.1, shall be packed in a weather resistant fiberboard box conforming to PPP-B-636, variety SW, grade V3c, type 1 load. The box shall be minimum cube to contain the unit packed breathing apparatus. Filler pads fabricated of the same material as the box, or filler pads specified in 5.2.1, shall be used as necessary to ensure a tight pack. The box shall be closed in accordance with the Method V closure requirements and reinforced using nonmetallic strapping as tape banding in accordance with the reinforcing requirements in the appendix to PPP-B-636.

5.3 Marking. In addition to the standard interior and exterior markings required by MIL-STD-129, the following markings shall be added:

(a) Shelf life code markings shall be applied to the packs and overpacks.
(b) Hazardous materials markings shall be applied for all modes of transportation as indicated:

(1) Proper Shipping Name: Oxidizing substances, n.o.s. (Potassium Super Oxide and Sodium Chlorate) UN 1479
(2) Hazard Class Label: Oxidizer"

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6.2(b): Delete in its entirety and substitute the following:

"(b) Level of packaging required"

6.3, lines 2 and 3: Delete "Federal Stock Number 4240-678-5263" and substitute "National Stock Number 4240-00-678-5263".

6.4: Delete in its entirety.

6.5: Delete in its entirety.

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Preparing activity:

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